Running the CalGames Content Desk

A complete setup and operating guide for volunteers

CalGames 2026 · Woodside High School · October 16-18 · Western Region Robotics Forum

The desk's front page and QUICK-START.pdf (one page; QUICK-START.md is the same text) are all a volunteer needs. The desk finds the field and the video switcher by itself, the front page says what it found and what to do next, and the desk console's doors check names anything still wrong, with what to do. Everything below is reference, for looking one thing up. Nobody needs to read it to run the show.

Read this first

This guide assumes you have never run this program, never used OBS, and never touched a video switcher. That is the expected starting point. Nothing here requires you to type commands, edit code, or understand how any of it works.

You do not need to read any of it to run the show: the quick start and the desk's own front page cover that. If you want the background to your job, find it below and read that chapter.

IF YOU ARE…	READ CHAPTERS
The AV coordinator (you run the desk, the screens and the sound)	**Start here**, then 2, 4 (*Without OBS* first), 5, 6, 10, 11
An emcee (you announce, host trivia, read shout-outs)	**Start here**, then 8, 10
The only volunteer	1, 1a, 2, 6
Setting up on Friday	1, 2, 3, 4, 5
Desk manager / producer	1, 2, 6, 7, 11
On-air talent (announcing, analysis, interviews)	1, 2, 8
Field tech	1, 2, 3, 9
Replay operator	1, 2, 7
Trivia host or arcade scorer	1, 2, 10
Anyone who just needs a screen on a TV	1, 2, 5

The one rule. If something looks wrong on the broadcast, the fix is almost never in this program. It is a person pressing the wrong button, and the person who notices should say so out loud. Nobody gets in trouble for calling out a wrong graphic. The show is a team sport.

Start here: the AV coordinator and the emcees

At CalGames 2026 two kinds of people run this: **the AV coordinator**, who owns the computer, the screens and the sound, and **the emcees**, who own the microphone. Everybody is doing this for the first time. That is fine: almost everything here runs by itself once it is set up, and setting it up is a list of steps with a check at the end of each one.

The goal for Friday is one thing: a practice match where every screen does the right thing on its own. Once that works, the rest of the weekend is mostly leaving it alone.

The weekend at a glance

From the event schedule (updated 2026-09-30). The rows in bold are the ones the desk crew plans around.

WHEN	WHAT	FOR THE DESK CREW
Fri 8:30 AM - 4:00 PM	Field and pit setup	Rack the desk laptop, do the first hour below
Fri 4:30 - 6:00 PM	Field open for calibration	Connect the field (step 7)
Fri 6:00 - 7:45 PM	**Practice matches**	**Your test. Run the practice checklist below**
Sat 8:00 AM	Pits and arena open	Desk on, doors check green
Sat 9:45 AM	Opening ceremonies	
Sat 10:00 AM - 12:00 PM	Qualification matches	
Sat 12:00 - 1:00 PM	Lunch	Shout-outs and thank-you slides rotate
Sat 1:00 - 5:30 PM	Qualification matches	
Sat 5:30 - 5:45 PM	**Awards ceremony**	**JA hands over the awards code**
Sun 8:40 AM	Opening ceremony	
Sun 9:00 AM - 12:30 PM	Qualification matches	
Sun ~12:30 - 1:00 PM	**Alliance selection**	**The selection board goes up by itself** when the scorekeeper puts Alliance Selection on the arena's audience display (if you are holding a screen, press Selection), and gives way to Match 1's overview when Match 1 loads
Sun 2:00 - 5:00 PM	**Playoffs and awards**	**Second awards ceremony**

Three things that are true at CalGames 2026

- **There are 38 teams, and three of them are second robots.** The field calls them 9999, 9998 and 9997. Every screen prints them as **100B** (The Wildhats), **6036B** (Peninsula Robotics) and **4159B** (CardinalBotics). If you type a team into the "when do we play" page, 100B works.

- **Queuing runs on FRC Nexus** (frc.nexus/cacg). The desk can show who is being called to queue on every side screen, but only with an API key: get one at frc.nexus/api and paste it on **Event settings** (`/s/setup`), under **Connections**, **FRC Nexus API key**, then **Save connections**. It takes effect at once. Without a key everything still works; the desk works out the queue from the field instead. That needs the field connected (step 7): with **neither** the key nor the field, the side screens and the "when do we play" page have no queue to show. Either way, the match on the field and the next one to start are always at the top of the on-deck list, even after a queuer has marked the next match "On field" in Nexus.
- **Every team's pit is known** (Aisle 1 is pits A and B, Aisle 2 is C and D, Aisle 3 is E and F). The "when do we play" page shows a followed team's pit.

The AV coordinator's first hour

Do these in order. Each step ends with **You should see**; do not move on until you do.

1. **Start the desk.** Plug the desk laptop into power and into the field network (the network switch Cheesy Arena and the video switcher are on). Copy `CalGamesContentDesk.exe` to it and double-click it. Nothing appears for up to a minute the first time (it is unpacking). Then your web browser opens by itself.
   You should see *The desk is running*. *Pick what this computer is for*., a row of big buttons, and the **Start here** box under them. A red error box instead says what is wrong and that nothing is running. If Windows asks about its firewall (it names **Node.js JavaScript Runtime**), choose **Allow access** and tick every network type (chapter 2, step 2): otherwise no other screen in the building can reach the desk.
2. **Write the address and the PIN on the whiteboard.** Both are in the **Start here** box. Every other screen in the building needs the address; the people at the consoles need the PIN.
   You should see an address like `http://172.22.0.23:8720/`. If the box says the desk has no address, the laptop is not on the event network yet: plug in the network cable, wait a few seconds, and reload the page. No restart is needed.
3. **Prove the screens work before anything else is connected.** In the **Start here** box, click **Fill the screens with sample data**, sign in with the PIN, and press the button. Then, in another tab, open the address with `s/watch` on the end and click **Program**. (Once the desk has found the field it refuses, because real matches are coming: skip this step then.)
   You should see a score bar with two outlined numbers at the bottom, on the purple CalGames backdrop, and **100B** among the red teams. Outlined means "a guess", which is right: this is sample data. Everything sample says SAMPLE on it, nothing is saved, and it **clears itself the moment a real match loads**. Two sample teams get a drawn SAMPLE robot in the photo slot; a real photo already uploaded for either comes back when the sample clears. A team that has said no to its photo gets no drawing either, and saying no on the Team media page takes the drawing off at once.

4. **Put the graphics into the switcher, and the show on the projector.** On the computer that feeds the video switcher's overlay input (the one that showed Cheesy Arena's score overlay), open the address with `s/program` on the end and press **F11**. On the computer whose HDMI cable goes to the projector, open it with `s/watch` on the end, click **Program**, press **F11**. Chapter 4 has the details, and OBS if this event streams through it.
   You should see the score bar from step 3: on black on the switcher's input, on the purple backdrop on the projector. Reload the front page on the desk laptop: its **Screens** line names each computer showing the graphics and says what to change there, if anything (a window that is not full screen, or a resolution the switcher does not run).
5. **Put the venue screens up.** On each TV, open the whiteboard address with `s/watch` on the end and click **Side screen** (*For the TVs beside the field*) or **When do we play?** (*For pit TVs*). Press F11.
   You should see the on-deck list and rankings, changing by themselves every fifteen seconds or so. Leave them alone from here on. If a TV cannot open the address at all while the desk laptop can, it is the firewall (step 1) or a Wi-Fi that keeps devices apart: chapter 11.
6. **Put the music on the music machine, not the stream machine.** On the laptop wired to the hall's speakers, open the address and pick **House audio player**. Chapter 10 explains why it must be a different computer.
   You should see the player page. Press **Turn this machine on as the house player**, then use **Add a clip** on that page to upload one walk-up or stinger sound. Press that clip's button on the desk console and hear it in the hall but **not** on the stream (listen to the stream on a phone with headphones). The desk's **Play music** button needs a Spotify setup done off-site by the content desk lead; without it, only clips play.
7. **Check the field and the switcher.** With the laptop on the field network (step 1; ask the field tech, the FTA, for a port), the desk finds the field by itself, looking every 30 seconds until it does, and finds the video switcher on the same network; nothing needs restarting. If you cannot get on it Friday, skip this: the desk runs without the field and you type scores.
   You should see *Field: connected* on the front page (reload it), and a **Switcher** line naming the ATEM. The desk console's **Doors check** reads **Field bridge: OK**. The dot next to *linked* at the top of the console is NOT the field: it only shows that the console can reach the desk, and it is green with no field at all. Still *searching* once the field is up: chapter 9.
8. **Print the QR posters.** In **Start here**, click **print the QR posters**. Tape them at the doors, the stands and the pits. They send people to the shout-out form, the trivia game and the "when do we play" page.
9. **Change the three codes** before doors (chapter 2). Practice works on the shipped ones; the event must not.

Testing during practice matches

Practice matches are your rehearsal. You do not need to clear the sample data first: it disappears from every screen by itself when the first real match loads. (The **Clear sample data** button on the desk console does the same by hand.) With the field connected, watch one match and tick these off:

- [ ] When the match loads, the program shows both alliances (the overview).
- [ ] When the match starts, it switches to the score bar by itself, and the clock counts down.

☐ The numbers are **solid** (not outlined) when the field is connected.

☐ When the score posts, the program shows the final score by itself, and the next match's overview follows once the field's own audience screen moves off the score (about 20 seconds, never more than a minute).

☐ The side screens show the next matches, without anyone touching them. On Friday those are the next **practice** matches; Saturday's matches take over once practice is done. If the field crew runs the field's Test Match to check a robot, nothing on the desk moves: that is correct. If the practice schedule was made with the wrong date, practice still stays on the side screens while practice matches are being played; a desk restarted in the middle of that shows Saturday's matches until the next practice match starts.

☐ The music ducks when the emcee talks, if you armed that cue (chapter 6, *Show automation*).

☐ The projector shows the match full screen. With OBS: it is recording, and the stream (on a phone) has no music.

☐ The **Doors check** on the desk console is green (chapter 3).

☐ When 100B, 6036B or 4159B plays, the screens say so, not 9999.

If all nine pass, you are ready. If one fails, chapter 11 lists the usual cause.

The emcee's card

You need one thing: a tablet or phone open on **Talent view** (the address, then *Talent view*, then the PIN). It shows the teams on the field, their ranks, how close each alliance is to its bonus points, and how to say team names. Chapter 8 explains every line of it.

Things the emcees usually do, and where each one lives:

YOU WANT TO…	DO THIS
Know who is playing next	Talent view, or look at any side screen
Run crowd trivia between matches	**Trivia host** page: one button, press it when its label says what you want next. Chapter 10
Read out shout-outs	The approved ones rotate on the side screens. The AV coordinator approves them on the desk console
Give out an award	The AV coordinator presses Show, you read, they press Reveal when you say the name. Chapter 6, *Awards*
Explain a delay	Ask the AV coordinator for the delay card, then talk. An explained pause reads as process

A number with an outline around it means the desk is guessing. Do not read an outlined score as final; wait for it to go solid.

What runs by itself, and what to leave alone

By itself, all weekend, with nobody touching it: the program switching screens to follow the match, the score bar, the side screens' rotation, the on-deck list, the "when do we play" page, the thank-you slides for the

WRRF Board, the WRRF Interns, the CalGames Planning Committee, the volunteers and Woodside High School, and the shout-out QR code on the side screens.

Needs a person: approving shout-outs (they wait in a queue until somebody reads them), the awards ceremony, and trivia if you want to run it.

Leave alone: the desk laptop's lid and power cable, the OBS settings once they work, and any screen that is already showing the right thing. Nothing gets better for being clicked, and every page reconnects on its own if the network blips. While the desk runs it keeps the laptop from going to sleep by itself, but closing the lid or pressing the power button still sleeps it: keep the lid open (or set closing the lid to Do nothing in Windows Settings, Power) and keep it plugged in. The display may still turn off; that is fine.

If anything goes wrong mid-event, double-click the exe again. If the desk is still running it just reopens it in the browser; if it stopped, it starts again and the day comes back exactly as it was: scores, cards, the run of show. Every screen in the building reconnects by itself. Consoles that need the PIN (the desk, talent view, replay, telestrator pad, arcade, trivia host, ref review, cards and the music machine) go to the sign-in page by themselves after a restart: type the PIN and each comes back to the page it was on. The on-air overlays and the venue TVs never ask. Chapter 11 has the rest.

Checking it works before the event

Start the desk with nothing attached and most screens are empty, because there is no field, no match and no event yet. That is correct, and it is also useless for finding out whether the thing works.

So: open the desk console, switch to **Everything** at the top, find **Sample data**, and press **Fill the screens with sample data**. (The Start here box's link to it opens Everything by itself.)

Every screen then has something on it, and you can walk the buttons in the console's **Main screen (program)** row (Pre-match, Match, Final, Analysis, Arcade, Selection, Explainer, Sponsor) and see each one. The side screens, the pit monitor and the phone page fill in too. **When you have finished, press Auto.** Every screen button takes the program screen and holds it there until someone presses Auto; until then the program will not follow the matches, and the practice checklist fails for no visible reason.

Things to know:

- **All of it says SAMPLE.** The teams are real numbers but the slide, the sponsor, the people on camera and the award all say so on their face. If any of it ever reaches a projector by accident, the room can tell.
- **Scores draw outlined, not solid.** That is the desk's way of saying a number is a guess. It is how you can tell a sample apart from the field at a glance, and it is why you can leave this up while you set a monitor.
- **Nothing is saved.** The sample slides, people and robot photos exist only while the desk is running. Restart and they are gone, and the real event files are untouched.
- **It refuses once the desk has found the field**, answering or not: that desk is at an event, and invented teams have no business on a screen.

- **The delay card, the safety message and the countdown are not filled in.** Each of those covers the whole screen, which would hide what you came to look at. Press them yourself from the console when you want to see them.
- **No award winner is ever shown.** That is the one thing the desk keeps secret until the reveal, and a sample does not get an exception.

`START-PRACTICE.cmd` does the same filling in and then runs a match on a loop, which is the better choice if you want to watch the clock and the score move.

Chapter 1 · What this thing is

The content desk is one program that runs on one laptop. Everything else (every screen, every control panel, every phone) is just a web page pointed at that laptop.

That is the single most useful idea in this guide, so it is worth saying twice:

> **There is one program. Everything else is a browser tab.**

So there is nothing to install on the pit monitors, nothing to install on the announcer's tablet, and nothing to install on anyone's phone. You type an address, and a screen appears.

What it does

- Draws the **scoreboard and graphics** that go on the stream and on the TVs in the hall.
- Tells the **announcer** who is playing, what their record is, and what they need to do to earn a ranking point. In words, not jargon.
- Cuts **replays** and **match videos**, and uploads them.
- Runs the **crowd trivia** and the **arcade** side tournament between matches.
- Runs the **house music** in the hall.
- Answers **"when does my team play?"** on any parent's phone.

The three jobs

Three people is the comfortable shape of this show, and the rest of this guide is written for it. It is not a floor: one person can run it, and chapter 1a is how. Read this section anyway, because it says what each job is for, and a one-person crew is doing all three.

Desk manager (producer). Sits at the desk laptop. Drives the graphics, switches cameras, keeps the show moving. If there is only one person who knows how the desk works, it is this one.

On-air talent. Announces, analyses, does interviews. Works off a tablet showing the talent view, which tells them everything they need to say.

Field tech. Owns the connections: is the desk talking to the field, is OBS up, is the stream live, is the recording running. Checks the pre-show list in chapter 3 and fixes what it flags.

Adding people. The jobs above are the first three. After them, in order: a **dedicated switcher** (so the desk manager stops cutting cameras), a **second talent** (so the announcer has somebody to talk to), a **replay operator**, then the trivia host and the arcade scorer.

Each one takes over a console that already exists, so nobody has to be retrained and nothing has to be reconfigured: hand them the address, give them the event PIN, and they open a page. A volunteer who turns up at lunchtime can be useful within five minutes.

Chapter 1a · If you are the only one here

It happens. Two people said yes and neither arrived, or they are both in the pits fixing a robot. This chapter is the whole show, run by one person, and it is deliberately short because you are busy.

The show still works. Nothing in this program needs a second operator. Every automatic behaviour the desk has exists for exactly this: the screens follow the match on their own, the score bar follows the field, the side screens rotate by themselves, and the "when does my team play?" page answers parents without you.

Do these four things before doors.

1. **Arm every cue.** Open **Show automation** on the desk console (press **Everything** at the top first: the Simple view leaves it out) and press **Arm every cue**. This is the opposite of the advice in chapter 6, which tells you to watch each cue be right first, and that advice assumes three people. Alone, you want the desk cutting cameras and ducking music by itself, because the alternative is that nobody does it. You can still take anything by hand. They stay armed through a restart.
2. **Put the side screens and the pit TVs on first.** They run unattended all day. Chapter 5.
3. **Run the doors check** and fix what it flags, because nobody else will notice later. Chapter 3.
4. **Decide what you are not doing** (below) and do not open those pages.

What to run, and what to leave shut.

OPEN IT	LEAVE IT SHUT
The desk console	The replay console
The talent view, if you are announcing	The telestrator pad
	The trivia host and arcade consoles
	The team media page

Leaving a console shut costs nothing. Nothing breaks, nothing queues up waiting, and no screen goes blank because a page is not open: every screen reads the desk, and the desk is running. The replay console in particular is pure upside when staffed and pure distraction when not.

The one thing to plan for: the awards ceremony. This is the only part of the weekend where being alone is genuinely harder, because the Judge Advisor has to hand you their code and the ceremony has a rhythm you cannot pause. Read chapter 6's awards section before doors, not during. Stage nothing yourself: the JA loads the winners from their own tablet, and all you do is press **Show** and then **Reveal** when the announcer says the winner's name.

Things that will feel wrong, and are fine.

- Nobody is cutting cameras between matches. The cues are. Let them.

- The announcer is you, and you are also driving. Put the delay card up (chapter 6) and take ten seconds. An explained pause reads as process.
- You will miss a replay. There is no replay operator. This costs the show nothing that the audience will notice.

If a second person turns up, hand them the replay console first: it is the job with the best ratio of value to instruction, and chapter 7 explains it in two minutes. After that, the talent view.

Chapter 2 · Starting the desk

Step 1: Copy one file

Copy `CalGamesContentDesk.exe` onto the desk laptop. Put it anywhere: the Desktop is fine.

That is the whole installation. There is no installer, no admin password, and nothing goes into Program Files.

Step 2: Double-click it

Nothing appears straight away. The first time, the desk spends up to a minute unpacking itself; after that it starts in a few seconds. Then **your web browser opens on its own**, on the desk's front page. That is the whole interface: there is no other window, on purpose (step 4).

If something is wrong, a red error box pops up and says what. It also says nothing is running, so it is always safe to double-click again.

Windows will probably also ask about its **firewall**, once or twice (once for the desk, once for the part of it that serves the screens; the box names **Node.js JavaScript Runtime**). Choose **Allow access** and tick every kind of network it offers. If it asks for an administrator, fetch one: without this, the desk works on this laptop and nowhere else, and every TV, tablet and phone in the building just fails to load the address. Missed the box? Search Start for **Allow an app through Windows Firewall**, press **Change settings**, and tick **Node.js JavaScript Runtime**.

Windows may show a blue **"Windows protected your PC"** box. This is normal for any program that has not paid for a code-signing certificate. Click **More info**, then **Run anyway**. If you are not comfortable doing that, get the content desk lead. Do not click past a security warning on somebody's word alone, including this document's.

Step 3: Read the address

The front page opens with *The desk is running. Pick what this computer is for.*, a short status (the field, the switcher, the screens and OBS, as of when the page loaded: reload to check again), and big buttons: **This laptop runs the show**, **Put the graphics into the video switcher**, **Put the show on a projector or TV**, **Announcer tablet** and **Judge Advisor**. The **Start here** box is under those buttons. On the desk laptop it shows something like:

```
 This desk on the event network: http://172.22.0.23:8720/
Desk PIN: 0864
```

Under it, **Something not working?** answers the common problems in a line or two each, and two folded sections hold every screen with its address and a note for anyone using OBS.

Write that address on the whiteboard. Every screen in the building is going to need it. Write the PIN there too. It is a door code, not a secret; roughly eight people need it.

The desk actually has **three codes**, one per job, and only the first goes on the whiteboard:

CODE	WHO HOLDS IT	OPENS	SHIPPED DEFAULT
Desk PIN	The desk crew	Every operator console	`0864`
Awards code	The Judge Advisor	`/s/awards`, winner entry and the reveal	`1357`
Settings code	The content lead	`/s/setup`, the event's content	`4567`

The defaults exist so a practice run works out of the box. **Change all three before the event.** Anyone holding this handbook knows them. The **Start here** box and the doors check's **Access codes** row name every one still in use, on the desk laptop only: no other computer is told which shipped codes still work.

Changing them is the content desk lead's job on Friday, not the field tech's on Saturday morning, and it needs no file and no restart. In the **Start here** box, click **Change the codes**: that is **Event settings** (`/s/setup`), which opens with the settings code. Under **Access codes**, each code has its own box: the current code, the new one twice, and a button. A change works at once, and anyone using the old code types the new one. The desk PIN can only be changed from the desk laptop itself. The Judge Advisor can also change the awards code on their own page (`/s/awards`): tell them the new one if you change it for them. Each code is 4 to 8 digits, and no two can be the same. No code lives in `config.json` any more: the desk keeps what Event settings saves, and it outlasts a restart and a new copy of the exe.

If the awards code is the one that slips, that is the expensive one. `/s/awards` opens its sign-in box for anyone on the venue Wi-Fi, so a default awards code means the whole building can read the winners before the ceremony.

The numbers will be different at your event. Use what your browser shows, not what is printed here.

The PIN only appears on the desk laptop itself. The same page opened on any other device on the network shows the address alone, because that page is open to everyone on the venue Wi-Fi.

Lost the browser tab? Double-click the exe again. If the desk is already running it just opens it again; it never stops a running show.

Step 4: There is no window, and that is deliberate

Nothing stays on screen after the desk starts. Your browser opens on the front page, and that is the whole interface. There is no black window sitting on the taskbar.

That is on purpose. A window that runs the show is a window somebody closes: it looks like leftover clutter from an install that already finished, and one click on the X takes the broadcast off air in the middle of a match. Asking people to leave it open is not a control.

If you want to see what it is doing, open `desk-log.txt` in the desk folder. Everything the old window printed is in there, including the address. The PIN is not: it is on the **Start here** box on this laptop, so the log can be sent to whoever is helping without handing them the door code.

To stop the desk when the event is over, either:

- Open Task Manager (**Ctrl+Shift+Esc**), find **CalGames 2026 Content Desk**, select it, and press **End task**. Or
- Open PowerShell and type:

```
Stop-Process -Name CalGamesContentDesk -Force
```

Either one takes the desk down cleanly and releases the address, so the next start works. Both take more than one careless click, which is the point.

If it will not start

WHAT YOU SEE	WHAT TO DO
"Port already in use": an older copy is still running	Double-click the exe again: that reopens the running desk and stops nothing. For a genuinely fresh start, end it first (Task Manager, **CalGames 2026 Content Desk**, End task) and double-click again. The stop-or-open question only appears when you start it from a console window
A new copy of the exe changes nothing	The old desk is still running, and a double-click only reopens it. End it in Task Manager first, then double-click the new exe
It says the field was not found	Fine. The desk keeps looking by itself and connects when the field is up; see chapter 9.
A red box: *This copy of the desk is missing some of its own files*	Nothing on this computer needs fixing. Get a new copy of the exe from the event lead and double-click that
Nothing happens at all	The file did not finish copying. Copy it again.
A blue "Windows protected your PC" box	More info → Run anyway. See step 2.

Trying it at home first

You do not need a field, a camera, or an event to practice. When the desk finds no field it asks whether you want a **pretend match**; press **D** and every screen plays a full match on a loop with realistic scores.

After the first run, two files appear in the desk folder:

- **START-DESK.cmd**: the real thing.
- **START-PRACTICE.cmd**: the pretend match.

Both are double-click. Practice on the second one as much as you like; it cannot touch a real field or upload anything.

Chapter 3 · The pre-show check (field tech)

Open the desk address in a browser on the desk laptop and press **This laptop runs the show** (`/s/desk` , the desk console). It asks for the PIN once. The doors check is in both the Simple and the Everything view.

The desk console's header carries the **doors check** verdict at all times, next to the link light: *all good*, *needs a look*, or *something is broken*. Click it, or the **Doors check** chip in the jump strip, and it takes you to the check itself and opens the fold it lives in.

It answers one question: *is this thing ready?* Every row is green, amber or red, and every row also spells its level out as a word (**OK**, **WARNING**, **FAILED**, **UNKNOWN**, **OFF**), so the printed sheet carries the verdicts and nobody has to tell the colours apart.

If the verdict ever reads **not answering**, the check itself has stopped running, and the rows underneath go faint because they are the last thing it managed to say rather than the state of the building now. Press **Run the check**. If it stays that way the desk window has probably closed; chapter 2 starts it again.

Run it **at doors**, meaning when the audience is let in, not five minutes before the first match.

This is every row it can show, in the words the screen uses. **Stream health** only appears once OBS is connected, and **Match timing** only when the field disagrees with the desk:

CHECK	WHAT IT MEANS WHEN IT IS NOT GREEN	WHAT TO DO
Desk mode	Amber: the desk is in PRACTICE (started with `/rehearsal`, `START-PRACTICE.cmd` or `/demo`) or REPLAY, so nothing done on it becomes the day's history	End the desk in Task Manager and double-click the exe before doors
Settings file	Gray: there is no `config.json`, which is fine: the codes and connections are set on Event settings. Red: `config.json` has a typo near the line it names, so none of it is in use: any code set in it (not one set on Event settings) is back to the shipped default, and OBS is off. Amber, which does not hold the doors: a section is written twice (it names the section and its lines; only the last one counts), or a section is not a group in `{ }`, for example `"obs": true` (it names the section, which is ignored, so for `"obs"` OBS is off)	Red: fix that line (check the quotes and commas), save, restart the desk. Amber: merge the repeats, or write the section as a group copied from `config.example.json`, save, restart
Field bridge	The desk cannot see the scoring system. Gray and *searching*: it has not found the field yet, and keeps looking every 30 seconds. Gray and *not looking*: the desk was started with `/no-cheesy`. Amber, *not answering*: it found the field and lost it	Put the desk laptop on the field's network. Still searching once the field is up: type the field's address on Event settings, **Connections** (chapter 9)
Match timing	Amber: the field's match periods differ from the standard ones this desk is built for	Ask the scorekeeper to put the standard periods back, or expect the endgame chip and the countdown to be off by that much
Switcher	Gray: no video switcher found, which is fine on a setup without one. Amber: two overlays keyed on air at once; a match running with the graphics input not keyed; the switcher turned the desk down; an address on Event settings it cannot use; or the switcher lost	The row says what to do, in the switcher's own words where it has them. Chapter 9, *The video switcher*
Screens	Amber: no program page open anywhere; one in a window instead of full screen; one at a resolution or refresh rate the switcher does not run; one showing the see-through OBS version on white; one that cannot reach Cheesy Arena's own display, so it has no fallback; or one showing Cheesy's overlay because the desk or the field link was down a moment ago	The row names each computer and what to do there. Chapter 4, *Without OBS*
OBS	The desk cannot drive the camera switching. Gray means it is switched off, which is right for an event without OBS	Only if this event uses OBS: chapter 4, step 7 turns it on in `config.json`
Stream health	OBS is dropping frames on the way out: amber past one in a hundred, red past five	The uplink is struggling. Tell the content desk lead now, not when it gets worse
Recording	Nothing is being recorded, so there will be no replays or videos. Gray says which part is missing: ffmpeg not installed, or no cameras listed in `config.json`	Chapter 4, step 9, or accept a show without replays. Red with a camera listed: check the capture device is plugged in

CHECK	WHAT IT MEANS WHEN IT IS NOT GREEN	WHAT TO DO
Disk	Running out of room: amber under five hours of recording left, red under two	Now, not at lunch: empty the Recycle Bin and move other files off this drive. Then move the oldest camera files (named by date and time, practice first) out of the camera folders in the `rec` folder inside the desk folder. Never touch `rec/clips`, which holds the cut match videos
Event log	Red, and it holds the doors: the day is not being written down. The row names the error, usually a full disk. A restart now would rebuild the day only up to the moment it stopped	Free disk space (see Disk). The desk tries a fresh file by itself every 30 seconds and the row clears. Do not restart the desk until it is green
Publishing	YouTube is not set up	Videos will not upload. Talk to the content desk lead
Microphones	A mic is muted in OBS	Unmute it
House audio	The music machine is not connected, or two machines are both armed	Chapter 10
Coverage	A match has been played and no video was ever queued for it	Tell the content desk lead before the recording rolls off the disk
Access codes	At least one of the three codes is still the shipped default (amber; it names which, and only on the desk laptop), or the desk PIN is empty (red, and it blocks: every control is open to anyone on the venue Wi-Fi; gray instead if the desk only listens on this laptop, 127.0.0.1, where nobody else can reach it)	Read the two paragraphs below. Do not skip this one. If it is red, set a desk PIN on Event settings, on the desk laptop

A row can also be **grayed out**, which means that part is switched off rather than broken: no field connection, no recording set up, no stream started yet. Gray **Recording** is fine on a desk without the optional recording setup; **Field bridge** and **OBS** should not be gray by the time the doors open if you are using them.

Amber is not red. Amber usually means "this will work, but you should know". A missing Blue Alliance key is amber: videos still upload, they just do not get linked.

Access codes is the one amber you do not shrug at. If it is amber and the words *awards code* are in it, go and find the content desk lead. The shipped awards code is printed in this handbook and in the public code, and the Judge Advisor's page opens its sign-in box for anyone on the venue Wi-Fi. That amber means the building can read the winners before the ceremony. Chapter 2, step 3 says how to change it: a minute on Event settings, with no restart, so do it now rather than at four o'clock.

Everything red on this list will be noticed by somebody in the audience within an hour. That is the entire point of running it early.

Chapter 4 · The program screen: without OBS, or with it

The program screen is the match graphics: who is playing, the live score, the final score, the awards. Most venues no longer run OBS: they put the program on the projector, a TV or a video switcher straight from a browser, which is the first section below. From **Step 1: Install OBS** on is only for an event that streams through OBS.

Without OBS: a browser, full screen

1. On the computer whose HDMI cable goes to the projector or the TV, open a browser at the desk address with `s/watch` on the end, for example `http://172.22.0.23:8720/s/watch` (the front page's **Put the show on a projector or TV**). On the desk laptop itself, open `http://localhost:8720/s/watch` in a new window dragged onto the projector.
2. Click **Program**, the card that says *For the projector, or the stream*.
3. Press **F11** for full screen.

Until a match loads it shows the purple CalGames backdrop, **CALGAMES 2026** and one line saying what it is waiting for (*What the projector says*, at the end of this chapter). When the field loads a match the graphics take over by themselves, and once a match has loaded the line stays away.

The video switcher. On the venue kit a computer's HDMI goes into the switcher's overlay input, which the switcher keys over the cameras (DSK 1 on this kit); it used to show Cheesy Arena's score overlay. On that computer open `ADDRESS/s/program` and press F11 (the front page's **Put the graphics into the video switcher**). It is graphics on black, and the luma key drops the black out. While the desk works, its graphics replace Cheesy's overlay; when the desk cannot give a score, the same input shows Cheesy's by itself, so key only this one input. Use a luma key, not a chroma (green) key: the CalGames colors include a green and a blue. An input the switcher cuts to, rather than keys, takes `ADDRESS/s/watch?screen=program`, like a projector.

How the hand-off to Cheesy's overlay works. The program page keeps Cheesy Arena's own audience display loaded, unseen, behind the desk's graphics, straight from the field (so that computer must be on the field network). It shows Cheesy's in their place when the desk has not answered for about 3 seconds, or when the desk is up but has lost the field for more than 10 seconds (Cheesy has the real score then). It goes back to the desk's graphics once the desk and the field have both been answering for 5 seconds. It never reloads itself while the desk is away, because a reload then would be a light browser error page keyed onto the stream. Nobody presses anything. The doors check's **Screens** row says when a program page is showing Cheesy's, and warns ahead of time when that computer cannot reach Cheesy's display, while there is still time to move its cable. Do not also key Cheesy's overlay from its own computer: two overlays on air is the **Switcher** row's first warning. The desk finds the switcher by itself and only ever reads it (chapter 9, *The video switcher*).

Why a page can look white or black. `ADDRESS/s/program` opened by itself in a browser is graphics on black, for a luma key, with nothing on it until a match loads; the telestrator render and the arcade overlay are the same. Inside OBS, or with `?key=alpha` in the address, they are see-through, which a plain browser paints white. So a projector always goes through `/s/watch`. Software that layers pages without being OBS (vMix, Wirecast, CasparCG) needs `?key=alpha` in the address.

With OBS

OBS Studio is the free program that sends the video to YouTube: the desk draws the graphics, and OBS puts them over the camera picture. A setup made before this change works exactly as it did: a Browser Source pointed at `/s/program` is still see-through and never shows the lines in *What the projector says*. Skip to chapter 5 if this event does not stream through OBS, or if somebody has already set OBS up.

Step 1: Install OBS

Download OBS Studio from **obsproject.com** and install it. Take every default.

Step 2: Make the scenes

A "scene" in OBS is one arrangement of pictures: a camera, a graphic, a title. You switch between scenes to change what the audience sees.

Create six scenes with **exactly these names**. Capitalisation and underscores matter, because the desk switches them by name:

SCENE NAME	WHAT IT SHOWS
`CG_INTRO`	The opening title, before the day starts
`CG_MATCH`	The field camera, the main one
`CG_SCORE`	The field camera, for the final score reveal
`CG_REPLAY`	The replay playback
`CG_DESK`	The analysis desk camera
`CG_ARCADE`	The video game capture, for between matches

To add a scene: in the **Scenes** box at the bottom left, click +, type the name, press Enter.

If you get a name wrong, the desk's camera buttons will do nothing and nothing will explain why. Check the spelling twice.

Step 3: Add the match overlay

This is the important one. It is the scoreboard.

1. Select the **CG_MATCH** scene.
2. In the **Sources** box, click + → **Browser**.
3. Name it `Overlay` and click OK.
4. Fill in the box exactly like this:

FIELD	VALUE
URL	`http://localhost:8720/s/program`
Width	`1920`
Height	`1080`
Use custom frame rate	leave unticked
Shutdown source when not visible	**UNTICK THIS**
Refresh browser when scene becomes active	**UNTICK THIS**

Those last two matter. If either is ticked, the overlay reloads every time you cut back to the camera, and the audience watches the scoreboard flash and re-animate mid-match.

Use `localhost`, not the address on the whiteboard, for every OBS source on this laptop (the Overlay, the Telestrator, the arcade overlay): `localhost` always means this laptop, while the whiteboard address can change if a cable is moved or the network renews it, and an OBS source left on the old address goes blank on air. The whiteboard address is for the other screens.

1. Click OK. The scoreboard appears over your camera.
2. Drag it so it fills the frame exactly. Right-click it → **Transform** → **Fit to screen** does this in one step.

Step 4: Copy the overlay to the other scenes

The scoreboard belongs on the score scene and the replay scene too.

Do **not** add a second Browser Source: that runs a second copy of the overlay and doubles the load on the laptop. Instead:

1. Right-click the `Overlay` source → **Copy**.
2. Select `CG_SCORE`, right-click in the Sources box → **Paste (Reference)**.
3. Do the same for `CG_REPLAY` and `CG_DESK`. (`CG_INTRO` is only the pre-show title and carries no Overlay; the desk cuts away from it when a plate goes up.)

Every scene that can be on air while you use name cards, a delay card, a safety message, a slide or the awards needs the Overlay: it is the only thing that draws them. Shoot the awards ceremony on a scene that has it (`CG_MATCH` or `CG_DESK`).

Paste (Reference), not **Paste (Duplicate)**. A reference is the same overlay showing in two places. A duplicate is two overlays, and they will drift out of step with each other.

Step 4b: The replay player

Replays play through OBS itself, in the replay scene.

1. Select `CG_REPLAY`.
2. + → **Media Source**, name it exactly `CG Replay Clip`.
3. Tick **Local File** and leave the file empty: the desk loads each clip into it by name.
4. Tick **Restart playback when source becomes active** and **Close file when inactive**. Untick **Loop**. OK.
5. Drag it **below** the Overlay in the Sources list (the Telestrator, if you add one, goes above the Overlay).

Step 5: The telestrator (optional)

If your analyst is going to draw on replays:

1. Select `CG_REPLAY`.
2. + → **Browser**, name it `Telestrator`.
3. URL `http://localhost:8720/s/tele`, 1920 × 1080, same two boxes unticked.
4. Drag it **above** the Overlay in the Sources list.

Step 6: The arcade overlay (optional)

Same recipe on the `CG_ARCADE` scene, with URL `http://localhost:8720/s/arcade`.

Step 7: Let the desk drive OBS

This lets the desk manager cut cameras from the desk console instead of reaching for OBS.

1. In OBS: **Tools → WebSocket Server Settings**.
2. Tick **Enable WebSocket server**.
3. Click **Show Connect Info** and note the **Server Password**.
4. Click OK.

Then tell the desk about it, in `config.json` in the desk folder. There is no `config.json` until somebody makes one: in the desk folder (`Downloads\CalGames2026-ContentDesk`), copy `config.example.json`, name the copy `config.json`, and open it in Notepad. It already has an `"obs"` block, copied from the example: find it, change `false` to `true`, and put the password between the quotes, so it reads:

```
"obs": { "enabled": true, "host": "127.0.0.1:4455", "password": "THE SERVER PASSWORD" }
```

Do not add a second `"obs"` line: the later one wins, silently. Keep every comma, and type the quotes in Notepad rather than pasting them from a document (curly quotes break the file). If the file has a typo, the desk says so on the **Start here** box and ignores the whole file until it is fixed.

Save, and restart the desk (on this laptop, Event settings, **Restart the desk**; or Task Manager, **CalGames 2026 Content Desk**, End task, double-click the exe). The **Doors check** on the desk console shows OBS green once it is connected. OBS has to be on this same laptop.

You can run the entire show without this. The desk manager just switches scenes in OBS by hand instead, and everything else works identically, with one exception: **Take to program** on the replay console needs this. Without it the replay console says *NOT on air*, names the clip file, and tells you how to play it by hand (program screen on Blank, load the file into `CG Replay Clip`, cut to `CG_REPLAY`).

Step 8: Check it

Load a practice match (chapter 6) and look at your OBS preview. You should see the scoreboard over the camera, with the clock counting.

If the overlay is **black instead of transparent**, the Browser Source lost its transparency. Delete it and add it again. Do not add a Color Correction filter, which is the usual wrong fix.

Step 9: Recording, replays and match videos (optional)

Replays, **Take to program** and the automatic match videos all cut from a recording the desk makes of the program feed. The exe does not include the recorder's video tool, so this is a one-time setup on the desk laptop, and the show runs without it (no replays, no match videos):

1. Install ffmpeg: open PowerShell and run `winget install Gyan.FFmpeg`. Then restart the desk.
2. In OBS: **Start Virtual Camera** (bottom right). The desk records that.
3. In `config.json`, find `"recording"` and replace `"sources": []` with `"sources": [ { "id": "program", "label": "Program feed", "role": "program", "input": ["-f", "dshow", "-i", "video=OBS Virtual Camera"] } ]`
4. Restart the desk. The **Doors check** row **Recording** should read one source rolling. Leave OBS's Virtual Camera on all day.

A note for events with a hardware switcher

Covered under *Without OBS* at the start of this chapter: an input the switcher keys over the cameras takes `ADDRESS/s/program`, which is graphics on black by itself in a browser (`?key=luma` says so explicitly, and `?key=fill` puts the CalGames backdrop behind them instead); an input it cuts to takes `ADDRESS/s/watch?screen=program`. On the OBS path, leave `?key=` off.

What the projector says

THE LINE UNDER CALGAMES 2026, WORD FOR WORD	WHAT TO DO
Connecting to the desk at http://ADDRESS.	Wait a few seconds
No field connection. Matches can be run from the desk console.	The desk is not connected to the field. It finds the field by itself once the laptop is on the field network (chapter 9); meanwhile run matches with **Load a match by hand**
Connected to the field. Waiting for it to load a match (its Test Match does not count).	Ask the scorekeeper to load a practice, qualification or playoff match. The desk ignores the Test Match on purpose
In yellow: *Cannot reach the desk at http://ADDRESS. Check that the desk computer is on and the Content Desk is running.*	The desk laptop is off or asleep, the desk was ended, or this computer is on another network. It heals by itself
A second line, in yellow: *No picture from the field feed at HOST: check that the camera is on and its address is right. Until then the graphics show on this backdrop.*	Only with a field feed set. Fix the camera or its address (chapter 5); the video then appears by itself
The backdrop with no words	Somebody pressed **Blank**: press **Auto**

Chapter 5 · The screens around the hall

Every TV, monitor and phone in the building shows a web page. None of them need the PIN.

The easy way

On the computer plugged into the display, open a browser at the desk address with `s/watch` on the end, for example `http://172.22.0.23:8720/s/watch` (on the front page, that is **Put the show on a projector or TV**). The page is headed **Pick a screen**. Click the card for the display you are standing at: each one starts with a gold *For …* line. Press **F11** for full screen.

That is it. The page remembers full screen, so if the monitor loses power and comes back, F11 once and it is right again.

The screens, and where each one belongs

In the same order as the cards on **Pick a screen**:

FOR	SCREEN	ADDRESS	WHAT IT SHOWS
The projector, or the stream	**Program**	`/s/watch?screen=program`	The main show picture: who is playing, the live score, the final score. It changes on its own as the field runs
The TVs beside the field	**Side screen**	`/s/watch?screen=side`	Who is on deck and the current rankings, rotating. Built to be read from across the room
Pit TVs	**When do we play?**	`/s/watch?screen=next`	When each team plays next, with start times that follow the field running early or late. Also the QR code on the side screens
The trivia monitor	**Trivia**	`/s/watch?screen=trivia`	The crowd game. Shows the join code between questions
The arcade TV	**Arcade**	`/s/watch?screen=arcade`	Scores from the side tournament, for a TV beside the game consoles

With no field feed set, Program and Arcade fill the whole display, edge to edge on any screen shape, with the CalGames backdrop. Until a match loads, Program shows the line from chapter 4's *What the projector says*.

Side screens specifically

The side screens are the ones most people in the hall will actually look at. They rotate on their own between "who plays next" and the rankings, with no operator.

They are designed to be read from thirty feet away, which means the type is large and there is not much on screen at once. That is deliberate. Do not try to fit more on.

If a side screen goes blank or shows "reconnecting", the desk laptop went away. The screen recovers by itself when the desk comes back; you do not need to touch the monitor.

Checking a display with the test card

Before trusting any TV, projector, or LED wall, open `ADDRESS/s/testcard` on it, full screen. Four looks and you are done:

- **All four corner marks visible?** If not, the display is cropping the edges, and it will crop the scoreboard the same way.
- **Is the circle round?** An ellipse means the picture is stretched.
- **Can you tell the dark gray steps apart?** If the left end of the gray ramp merges into black, the display will crush the broadcast's purple too.
- **Is the clock ticking?** A frozen clock means a frozen signal.

The dashed rectangles are the exact safe areas the graphics keep to, so whatever the display does to them, it does to the score bar.

The field feed behind the program screen

`/s/watch?screen=program` shows the overlay on top of a live picture of the field. That picture comes from a setting the content desk lead fills in on the event settings page, not on the desk console: `ADDRESS/s/setup`, then **Screens and stream**, then **Field feed URL**. That page takes the **settings code**, which is not the desk PIN (chapter 6, Event settings). If the box is blank you get the overlay on the CalGames backdrop instead, which looks intentional and is fine. For one display only, add `&feed=` and the feed's address to that display's address.

With a feed set, the display waits up to about four seconds for the camera's first picture before it decides; a working camera usually answers in under a second, and then the video fills the display with the score over it.

- **The camera is off, on another network, or its address is wrong.** The display shows the program on the CalGames backdrop, with a second line in yellow: *No picture from the field feed at HOST: check that the camera is on and its address is right. Until then the graphics show on this backdrop.* It never goes black, and it names only the camera's host and port, never the rest of its address. It keeps asking every few seconds, and when the camera answers, the video appears under the score by itself (the graphics blink once). Nobody needs to touch the display.
- **A working camera drops in the middle of the show.** The score stays up on the CalGames backdrop instead of a black screen, with no words, and the video comes back by itself when the camera does.
- **Do not put a user name and password in the feed address** (`http://user:pass@camera/...`). Browsers refuse those, so the camera is never even asked. Set the camera to allow viewing without a login on the venue network, or use a capture box.
- **A YouTube feed** behaves as before: YouTube's own player says when the video is unavailable.

Chapter 6 · The desk console (desk manager)

Open `/s/desk`. Enter the PIN once (again only after a desk restart, when the console takes you to the sign-in page itself).

This is the main control panel and it is built to be driven by keyboard. The buttons all say what they do; this chapter covers the order you use them in.

The rhythm of a match

The field drives most of this by itself. What follows is what you do when the field connection is working, which is the normal case. Without OBS there is nothing to cut in the *You do this* column: the projector follows by itself.

WHEN	THE DESK DOES THIS BY ITSELF	YOU DO THIS
Next match is up (after the score has had its time)	Shows the alliance overview with robot photos	Cut to `CG_MATCH`
Field arms the match	Score bar comes in	Nothing
Match starts	Clock runs, scores update live	Watch the show, not the desk
Buzzer	Clock stops	Cut to a wide, or to the desk
Referees finish	Nothing	If a card was given, chapter 7
Score posts	Final score screen appears, and stays until the field's own audience screen moves off the score: about 20 s (30 s after a playoff match that is not a final), never more than a minute, and at once if the next match starts or the scorekeeper re-runs the match just played	Cut to `CG_SCORE`
Gap before the next match	Nothing	Sponsor, trivia, arcade, or an interview

If the field connection is down, you press those beats yourself: load the match with **Load a match by hand**, then the **Match steps** buttons near the top of the console are Preview, Arm, Start, End, Post score, Abort. The graphics behave identically.

Simple and Everything. The console opens in **Simple**, only what a match day needs, plus *Field down? Run matches by hand* while the field is not connected. **Everything** shows every panel; with no OBS set up, its cameras fold behind **Using OBS? Show the camera controls**. Every key works in both.

Keyboard shortcuts

The console is built to be driven by keyboard, because reaching for a mouse during a match is how you miss the thing you were watching for.

The match beats, along the number row:

KEY	DOES
`1`	Preview: alliance overview
`2`	Arm: score bar in
`3`	Start the match clock
`4`	End: buzzer
`5`	Post the score
`0`	Abort: the clock stops and the program goes back to this match's overview

Holding a key down does its action once. To do it again, let go and press it again.

Everything else worth learning on day one:

KEY	DOES
`Space`	**Mark a replay moment.** The one to learn first
`T`	Show the lower third (the name strip)
`Y`	Hide it
`B`	Blank the program screen

The next keys are only for scoring by hand, when the field connection is down. Red on the left of the keyboard, blue below it, the way they sit on the field:

KEY	RED	KEY	BLUE
`Q`	+1 fuel	`A`	+1 fuel
`W`	+5 fuel	`S`	+5 fuel
`E`	+10 tower	`D`	+10 tower

Anything you type this way renders **outlined** on air, so the audience is never shown a guess dressed as a result. See chapter 8.

Camera scenes are buttons on the console rather than keys, because cutting to the wrong camera by fumbling a key is worse than reaching for a mouse.

The console never steals a key while you are typing in a box, and it leaves Ctrl and Alt combinations to the browser: Ctrl+0 resets zoom rather than aborting the live match.

Putting people on camera

When somebody sits down for analysis or an interview:

1. In the **On camera** section (*Names for the people on camera*, in Everything), tick their name in **Saved profiles**. If they are new, type their name and role and press **Add**.
2. Check the order in **On air, in screen order**. Seat 1 is the audience's **left**. Use the arrows to move people.
3. Press **Put on air**.

Their name cards appear across the bottom of the screen, spaced evenly.

If any of them is a student, tick "Student". Their card then shows a first name and a family initial ("ALEX R.") instead of a full name. Everyone in the hall still knows exactly who they are; a search engine does not, and the video is public forever. Adults doing a job under their own name (mentors, coaches, the commentators) stay in full.

If you put students on with no adult on the panel, the console tells you so. Check the shot before you cut to it: YouTube can disable chat or take a stream down over minors on camera without a visible adult, and getting that wrong ends the stream, not the clip.

Press **Hide** when they get up.

A lower third (the single name strip, `T`) leaves by itself after its time on screen unless **Pin (stays up)** was ticked, and stays gone when OBS or a screen reloads the overlay. A pinned one stays until you press **Hide** (`Y`).

Status cards and safety messages

Two different things, deliberately.

- **Status card**: "Field delay", "Back in ten minutes". Explains a pause. It **stays up until you press Hide card**: only the field's own break card clears itself. That one names the break ("Awards Break", "Lunch", or "Field timeout" for a plain timeout) with its end time, goes when the field comes back or the next match starts, and on the stream steps aside while an award or slide plate is up. The Delay card section tells you on screen whenever one is live, and the jump strip has a **Delay card** chip to get back to it. The five buttons cover almost everything. If you type your own, keep it short: the card holds about ninety characters, the box stops you at that, and it counts the last thirty down beside the field so you can see the end coming. Fill in **Back in** and the time appears on its own line under the message, which is the part the hall actually wants. Once any card's time has passed (an Awards Break often runs over), every screen says "Back soon" instead of a time that has gone, so update or hide the card if the pause runs long.
- **Safety message**: covers every screen in the building, and stays until you clear it. Type it and press **Put on every screen**; **Clear** takes it down. This is for an evacuation or a hold. It is not for a long queue.

Both go to the program, the side screens and every phone at once. The side screens show a safety message on their next frame; phone pages pick it up within 5 seconds (When do we play) or 8 (shout-outs, crowd trivia); the trivia and arcade screens show the same safety plate, so a trivia or arcade monitor and the `CG_ARCADE` scene carry it too. When you send a safety message the console goes red so that whoever walks up next can see one is live. A restart does not lose it: it comes back with the desk.

Awards

Awards involve a second code, and it is not yours.

CalGames runs two ceremonies. Saturday presents the community and volunteer awards (Directors', Volunteer of the Year, Mentor of the Year). Sunday presents the team awards, nine of them, opening with the Founders' Award and closing with the Judges' Award. Both the Judge Advisor's page and your Awards section group by ceremony and number each award in running order, so you are only ever looking at tonight's list.

The **Judge Advisor holds the awards code**. All day, as judging wraps up, the JA loads each award's winner from their own page (`ADDRESS/s/awards` , which takes only their code, not the desk PIN). That page is also where the award list itself gets built: **Add an award** for each judged award in ceremony order, **Edit wording** to fix a typo, **Remove from the ceremony** if plans change. No file editing, ever. While that is happening, your Awards section shows only the lock and its code box; even the award titles wait behind it. That is deliberate. Nobody at the desk can see, type, or reveal a winner, so nobody at the desk can leak one, even by accident, even with a camera pointed at the console.

Right before the ceremony, the JA gives you the code. Type it into the Awards section's unlock box. Then, for each award:

1. Pick the award from the list. Ones marked **winner loaded** are ready: the JA's answer rides along invisibly.
2. Press **Show the award**. The title goes up with one line under it saying what the award is for. The GA reads the full definition aloud from the slides; the screen does not reprint it, because a paragraph on the big screen is a wall nobody in the hall reads.
3. On the GA's cue ("...and the winner is"), press **Reveal the winner**. The line comes off and the winner takes the frame.
4. Press **Clear** before the next one. Presented awards tick off the list, and the ceremony heading counts down ("4 of 9 presented").

The winner box on the desk is only for last-second corrections (the envelope on stage disagrees with what was loaded); leave it blank otherwise. The winner genuinely does not exist anywhere the audience can reach until Reveal, so there is nothing to leak. The lock means that now includes this console.

Each award carries two pieces of text, and the difference matters. The **full definition** is the committee's wording: what the GA reads aloud and what the projected slides show. The **on-air line** is the single line the broadcast puts under the title. The JA can edit both; leaving the on-air line blank falls back to the definition's first sentence, which is usually the right line anyway.

The desk ships with a default awards code (1357), so the lock exists from the first boot; change the code for the event, the same as the PIN. An event small enough that the producer IS the Judge Advisor can set `awards.pin` to empty in `config.json` (it applies only while nobody has changed the awards code on Event settings or the Judge Advisor's page), which removes the lock and runs awards on the desk PIN: then type the winner yourself before Show, and do not read it aloud to anyone standing behind the desk.

Slides and shout-outs

Slides are the between-matches cards: thank-yous for volunteer crews, session announcements, event info. The side screens rotate the whole deck on their own all day; **Next slide** steps through the same deck on the program when you want one on the big screen, and **Hide** takes it down again.

To add one during the event: pick a kind, type a title and lines (separate lines with `|`), press **Add**. It is saved immediately and survives a desk restart.

The thank-you slides are already loaded. The WRRF Board of Directors, the WRRF Interns, the CalGames Planning Committee, the volunteers and Woodside High School each have one, plus a slide inviting the crowd to send shout-outs. They rotate on the side screens from the first start. To name people (a setup crew, the planning committee by name), add a slide of your own.

The **shout-out queue** below it fills from `/s/gp`, a phone page where anyone in the building can thank a volunteer, a team, a mentor, the ref crew. They say **who it is for** and **what they did**; approved, it airs as *Shout-out to (who)* with the message and *From (name)* underneath.

People find the form three ways: the QR code in the corner of every side screen switches to it whenever a slide is up, the QR posters you printed (Start here, step 8), and the "Send a shout-out" slide in the rotation.

Nothing from that queue reaches any screen until you press Approve. Read each one; approve the good ones into the deck; reject the rest. Nobody is notified of a rejection. Check the queue between matches: it is the one part of the desk that waits for you.

The event timer

In Everything, under **Countdown on the side screens**: **Field setup 2:00** starts a countdown that takes over the side screens in digits readable from the far end of the gym; turn one side-screen TV to face the field and the drive teams have their setup clock. It clears itself the moment the match starts. The label and minutes boxes run any other countdown: meeting starts, doors, end of lunch.

Seeing what the screens look like, before there is a show

With no field and no match there is nothing to show: the program says so (chapter 4, *What the projector says*), and the other overlays are empty.

For setting a pit monitor up, aiming a projector, or showing somebody what the broadcast looks like, the desk console has a **Sample data** panel, in the **Everything** view under *Field down? Run matches by hand*. One press fills every screen with invented teams, scores, rankings and slides. It is for testing, not for the show:

- Everything it puts up is tagged as sample, so scores draw **outlined**, the same way a desk-typed guess does, and none of it can be published.
- The button refuses outright while the field bridge is running, because a desk talking to the field is a desk at an event.
- Load a real match, or restart the desk, and it is gone.

Show automation

In the Everything view, under **Show automation**, is a list of cues. A cue is the desk doing one thing by itself: cutting to the field when the match arms, dropping the music when the announcer opens up, putting the score up when a result posts.

Every cue ships switched off, and that is the right way to start. An off cue still watches the show and counts what it *would* have done, so the row tells you things like *off, would have run 4x*. Run a few matches by hand, read those counts, and arm the ones that were right every time. There is no prize for arming them all on Saturday morning.

- **Arm** lets one cue run by itself from then on. **Turn off** stops it.
- **Run now** fires a cue once, by hand, whatever it is set to.
- **Arm every cue** does the whole list at once, and asks first.

What you arm is remembered: a restart brings every cue back armed or off exactly as you left it. A desk started in practice mode starts with every cue off and remembers nothing you change in it.

You can always take anything by hand afterwards. Autopilot never cuts away from the field in the middle of a match. While an award or a slide is on the program, or a screen you took by hand, autopilot leaves the camera where you put it: On deck, Score reveal and Fill the gap all wait. The one exception is the field itself: when it goes ready or starts a match, Armed and Live still cut to `CG_MATCH`. Press Auto to hand the program back; automatic cuts resume as soon as the program moves off the plate, at the next match load. To take a plate down at once, press **Hide** for a slide and **Clear** for an award.

Plates on the stream: `CG_ARCADE` and `CG_INTRO` carry no Overlay, so an award, a slide, a sponsor card or the selection board put up while the stream is on one of them would never be seen. When you press Show the award, Reveal, show a slide or a sponsor card, or Selection while OBS is on either, the desk cuts to `CG_MATCH`, armed or not; on any other scene it does nothing. The same cut back for the selection board going up by itself also runs disarmed when Fill the gap made the cut to the arcade in that same gap, so with only Fill the gap armed the board still reaches the stream. In any other gap the board's automatic cut waits for this cue to be armed.

If OBS is not connected, every **Run now** still works: the graphics still switch, and only the camera cut is missing, which you take by hand in OBS. The panel says how many cues that applies to. The exception is **Take replay**: it runs whenever the replay console takes a clip, armed or not, and it needs OBS. With OBS down it refuses, and the replay console says why, instead of cutting to an empty replay scene. Autopilot never cuts away from a replay while it plays. **Fill the gap** fires only when a score has been up three minutes and no next match is loaded (the end of a session, or a stall), and never over an award, a slide, a held screen, the selection board or a

replay. The music cues and the end game chip never touch OBS anyway. Arming works for all of them either way, and an armed cue starts cutting cameras again the moment OBS comes back.

If a cue fails, its row says so in words and keeps saying so until it next succeeds. That row is the only place a failed cue is reported, which is the other reason to read this panel between matches.

Event settings: names, sponsors, and the run of show

Event content lives on its own page: `ADDRESS/s/setup`, behind the **settings code**, which is not the desk PIN and not the awards code. The content lead holds it. It is a before-doors tool: everything about the event that used to require editing a file lives there, saves the moment you press the section's Save button, takes effect immediately, and survives a restart. The desk console links to it from the Setup fold (in Everything).

- **Event**: the event name and year (stream titles use them), the TBA event key, and the results link that goes in video descriptions.
- **Bonus RP thresholds**: the offseason committee's numbers. Saving moves the badges and the talent view together, immediately.
- **Screens and stream**: the field feed URL behind the pit monitors, and the webcast URL registered on TBA.
- **Sponsor list**: name, tier, one line the announcer can read, and an optional logo path. Tier decides how often a card comes round, never how big it is drawn.
- **Run of show editor**: the day as a list. A matches block takes a match COUNT and times itself from measured pace; everything else takes minutes. Editing mid-day keeps the progress of every segment that kept its label.
- **Accessibility services**: each line is something a person in the building can act on, plus who to ask. Never list a service the event does not have.
- **Access codes**: the desk PIN, the settings code and the awards code, each changed with the current one (chapter 2, step 3). The desk PIN only on the desk laptop.
- **Connections**: **Field address (Cheesy Arena)**, **Video switcher address (ATEM)**, the switcher input carrying the desk's graphics, and the **FRC Nexus API key**. Leave the addresses blank and the desk looks by itself; type one only when it cannot find it. **Save connections** says which change, if any, waits for a restart.
- **Restart the desk**: on the desk laptop only, between matches. Every screen reconnects by itself and every console asks for its code again.

The codes, the Nexus key and the field and switcher addresses used to mean editing `config.json`; none of them does now. What stays in that file is the rest of the machine wiring (OBS, the camera list for recording) and upload credentials. If a section on that page asks for something you do not have, that is a question for whoever set the desk up, not a file for you to edit.

The awards list is the one content list NOT here: it belongs to the Judge Advisor's own page, behind the awards code (chapter 6, Awards).

Chapter 7 · Cards, replays, and the awkward moments

When a team gets a card

The field usually tells the desk, and a small mark appears next to that team's number: an outlined square for yellow, a filled block for red. Different *shapes*, not just different colours, so it survives a compressed stream.

For the announcement itself, use the **card call** screen. This puts a full-screen card up after the buzzer and before the score, so the announcer has something on air while they explain it.

1. Pick the team from the list: it only offers the six who just played.
2. Pick yellow or red.
3. Type the reason **in the words the announcer will say**. It goes on screen 200 pixels tall in front of the whole gym.
4. Press **Put the card up**.
5. Press **Clear** before the score reveal.

If the desk says the card is still up after you press Clear, press Clear again. It will tell you if it failed; it will not leave you guessing.

Replays

Press `Space` any time something interesting happens. You do not need to be exact: the mark lands a couple of seconds *before* you pressed, because everybody presses late.

The replay console (`/s/replay`) lists the marks. The desk has also marked things on its own: scoring bursts, lead changes, the end of auto.

To air one: click a mark (it frames 6 seconds before the moment and 4 after) or a spot on the timeline, then **Cut clip** (`Enter`). Half and quarter speed are ready in seconds. Then `T` or **Take to program**. *On program* means OBS is playing it, and the stream cuts back by itself when the clip ends: to where the show was before the first of any back-to-back replays, or to where autopilot wanted to go while the clip ran. If the switcher has already cut somewhere else in OBS during the clip, that cut stands and nothing cuts back. *NOT on air* says why, and how to air it by hand. While a replay plays, automatic cuts wait for it; the field arming takes the stream back at once. **Cut full match video** greys out while it cuts, and so does **Cut clip**: press it again after *Ready*. Holding a key down does its action once. The desk encodes at most two clips at a time and the rest wait their turn, so a replay cut can wait a little behind a full match video.

Surrogates

If a team is playing a surrogate match, one that does not count for their record, the graphics say so. Nothing for you to do; it is there so the audience is not told a team lost a match that was never theirs to lose.

Chapter 8 · The talent view (on-air talent)

Open `/s/talent` on a tablet. It asks for the PIN once, and again only if the desk restarts; it takes you to the sign-in page by itself.

Everything on this page is something you can say out loud. That is the design rule: no jargon, no raw numbers that need translating.

What is on it

- **Who is playing**, with team names, records, and ranks.
- **What each alliance needs for a ranking point**, in a sentence. Not "ENERGIZED: 87/100" but "thirteen more fuel for Energized".
- **Pronunciation notes**, where somebody has written one in.
- **Cards a team is carrying**, so you are not surprised.
- **Whether a team is a surrogate**, so you do not say the wrong thing about their record.
- **How the day is running**, ahead or behind schedule, in minutes.

A number with an outline around it

If a score renders as an **outline instead of solid**, it means the desk is *guessing*, usually because the field connection dropped and somebody at the desk is typing scores in by hand.

Do not read an outlined number as final. Say "unofficial" or wait. When the real score posts it goes solid.

This is the single most important thing on this page. The graphics will never present a guess as a fact, and neither should you.

Interviews

The desk manager puts your guests' names on screen. Give them a moment to do it before you throw to the interview: the cards animate in, and starting the question before the name lands means the audience never learns who is talking.

Chapter 9 · The field connection (field tech)

The desk can read the scoring system so the graphics follow the real match without anyone pressing anything.

It is read-only, by construction

The desk **cannot** start a match, abort a match, change a score, or touch the field in any way. That is not a policy: the connection is built so that those requests do not exist. You can hand this document to the FTA and the head referee and that sentence is the whole of it.

Setting it up

Usually nothing. The desk looks for the field by itself: the launcher looks once as it starts, and the desk keeps looking every 30 seconds until it finds the field, then connects on its own. Nothing needs restarting, so the desk can be started before the field network is plugged in. The front page says *Field: searching* until then.

If it still says *searching* once the field is up:

1. Check the desk laptop is on the **same network** as the scoring system. Event Wi-Fi and the field network are often not the same thing.
2. Check the scoring computer is on and its display is up.
3. Ask the FTA for the field's address and type it on **Event settings** (`/s/setup` , the settings code), under **Connections**, **Field address (Cheesy Arena)**, then **Save connections**. The desk looks there first, straight away.

If the front page says *not connected, and this desk is not looking for it*, the desk was started with `/no-cheesy` . End it in Task Manager and double-click the exe: a plain double-click always looks.

The video switcher

The desk reads the ATEM switcher, read-only like the field: it can see what is on program, which keys are on air and the video format, and it has no way to change any of it. It finds the switcher by itself, trying the field network (10.0.100.x, where the kit's ATEM lives at 10.0.100.40) and then this laptop's other networks, and the front page's **Switcher** line names it.

THE SWITCHER LINE SAYS	WHAT TO DO
No video switcher found (gray)	Nothing, on a setup without one. With one: check it is on and on the field network, or type its address under **Video switcher address (ATEM)** on Event settings
It would not let the desk connect (amber)	The switcher has no free connection. Close any copy of ATEM Software Control nobody is using; the desk tries again by itself
The address on Event settings cannot be used (amber)	Type four numbers with dots, such as 10.0.100.40, with no port, or clear the box to let the desk look
Lost the switcher (amber)	Check its power and network cable. The switcher keeps running the show either way

If the graphics reach the switcher on a known input, pick it under **Switcher input carrying the desk's graphics**: the doors check then warns when a match is running with that input not keyed on air.

If it stays down

The show runs anyway. Tell the desk manager, who switches to driving the match beats by hand (chapter 6). Everything looks identical to the audience except that scores are typed rather than read. Those render outlined, so nobody is misled.

This is worth rehearsing once on Friday. A field connection that drops during finals is a much better experience for everyone if the desk manager has done it before.

The other connections you own

THING	HOW YOU CHECK IT	IF IT IS DOWN
Video switcher	The front page's **Switcher** line, or the doors check	Chapter 9, *The video switcher*. The show runs either way
FRC Nexus	The side screens show who is called to queue	Paste the key on Event settings, **Connections**; it takes effect at once
OBS	The doors check, and the OBS window itself	Chapter 4
Stream	YouTube Studio shows it live	Restart the stream in OBS; the desk is unaffected
Recording	Doors check says Recording	Replays and videos stop; the live show is unaffected
Disk space	Doors check says Disk, in *hours*	Free space now

Chapter 10 · Trivia, the arcade, and the music

Crowd trivia

Open `/s/triviadesk`. The job is one button, and its label always tells you what the next press does: **Open question** puts it on the board and starts the clock, then it reads **Reveal** (with the seconds left, so you know when the window closes), then **Next**. Press it when it says the thing you want.

The crowd joins on their phones from the QR code the trivia screen shows. There are more than thirty questions loaded already, in rounds, so you can play a round after one match and pick up a different round after the next. Scores carry across the day.

Pick the winner asks the crowd who takes the match that is up. Queue it and open it before that match starts; once its score is posted it can no longer be queued. Reveal it any time after the score posts, even after the next match has loaded. In a playoff it pays the alliance the field declared the winner, tiebreaks included. If the field moves on without posting that match, the host console says so, and **Next** drops the round with nobody paid.

A round that has been played through will refuse to replay: that is on purpose, because replaying one pays everybody twice and quietly corrupts the leaderboard.

Names the crowd types are screened before they go on screen. Some will still get through that a reasonable person would not want on a projector; you can remove any name from the host console.

The arcade

Open `/s/arcadedesk`. Score the game sets by hand: two players head-to-head, or three-to-four free-for-all. The overlay shows the standings.

Each info box on the arcade overlay can be shown or hidden on its own, so you can clear the screen when the game itself is the thing to watch.

House music

This is the one part that runs on a different computer.

The music machine is whatever laptop is plugged into the hall's PA. Open `/s/house` **on that machine** and leave it open.

Why a separate machine: music played in the hall must not reach the stream. A recorded track on a YouTube video can get the whole archive muted or blocked, and the archive is what teams watch afterwards. Keeping the music on a machine that is not wired to the stream makes that structural rather than a thing somebody has to remember at 4pm on a Sunday.

The page refuses to run inside OBS for the same reason. If you see it refuse, that is it working.

The manager does all of this from the desk console, without leaving their seat: pause the music, resume it, duck it under an announcement, fire a team's walk-up song, or hand the room to the video game audio.

Chapter 11 · When something goes wrong

By how often it happens. Short answers: the front page's **Something not working?**, and QUICK-START.pdf.

WHAT YOU SEE	ALMOST CERTAINLY	FIX
A white or black page, nothing on it	An address for OBS or a switcher, on a projector. On the switcher's computer, black is right until a match loads	`ADDRESS/s/watch`, **Program**, F11. White on any computer: the front page's **Screens** line, on the desk laptop, names it and the address to open instead
The switcher's overlay input shows Cheesy Arena's own score, not the desk's	The desk stopped, or it lost the field for more than 10 seconds	Nothing: it goes back by itself 5 seconds after both answer. Find out why on the front page
Other screens cannot open the address at all, but the desk laptop can	Windows Firewall on the desk laptop said no, or the Wi-Fi keeps devices apart (common on guest networks)	Allow access (chapter 2, step 2), or have an administrator allow inbound TCP 8720; for Wi-Fi that isolates devices, use the wired network or ask the venue to turn isolation off
The program screen does not follow the match	A screen button was pressed, which holds the screen	Press **Auto** on the desk console
A screen says "reconnecting"	The desk laptop went to sleep, or the desk was ended	Wake it, or double-click the exe again. Screens recover on their own
The overlay in OBS is frozen	The Browser Source has "shutdown when not visible" ticked	Chapter 4, step 3
Scores are not updating	Field connection dropped	Chapter 9. Drive it by hand meanwhile
A score has an outline around it	The desk is guessing, and saying so	Nothing. It goes solid when the real score posts
Camera buttons do nothing	OBS is not connected, or a scene is misspelled	Chapter 4, steps 2 and 7
Music is coming out of the stream	The house page is on the wrong machine	Chapter 10: this one matters
No robot photos	None uploaded for those teams	Fine. It falls back to a gold team number
A wrong name on screen	Somebody typed it	Forget that profile, add it again spelled right, tick it, press Put on air
The final score sits for a full minute after every match	The field's automatic audience display is off (Cheesy, `/setup/settings`, **Enabled automated audience display**), so the desk waits out its one-minute limit	Ask the scorekeeper to tick it, or to press Intro on their audience display
The final score came off almost at once	The scorekeeper changed the field's audience screen, or started the next match	Nothing: the desk follows the field

The universal fix

Double-click the exe again. If the desk is still running, that simply reopens it in the browser, at once (before it unpacks anything or looks for the field). If you added an option such as `/pin` or `/cheesy-host`, a box says it was ignored because a desk is already running. If it really has stopped (the laptop restarted, or it crashed), that starts it, and it comes back in a few seconds remembering the day: cards (on Sunday, Saturday's qual cards too), which cues are armed, coverage, the run of show, sponsor counts, the day video chapters, quiz scores, the match pace, and any safety message, delay card, countdown or announcement that was up (a countdown that ran out meanwhile stays gone; the field's own break card keeps the field's end time, and comes down by itself if the break ended while the desk was down). Every screen in the building reconnects on its own. Consoles that need the PIN (the desk, talent view, replay, telestrator pad, arcade, trivia host, ref review, cards and the music machine) go to the sign-in page by themselves after a restart: type the PIN and each comes back to the page it was on. The on-air overlays and the venue TVs never ask. The music machine needs the PIN again too, and the music stays stopped until it gets it.

If the desk is running but misbehaving and you want a genuinely fresh start, press **Restart the desk** on Event settings, on the desk laptop: it comes back by itself with the same field, switcher and codes. Or end it: Task Manager (Ctrl+Shift+Esc), find **CalGames 2026 Content Desk**, End task, then double-click the exe.

You do not lose the day by restarting. That is deliberate, and it was built because laptops get their power cables kicked out.

When to escalate

Get the content desk lead for: anything involving credentials, uploads going to the wrong channel, or a message on screen that should not be there and will not clear. Everything else on this page is yours to fix.

Appendix A · Every screen, in one table

Replace `ADDRESS` with the address in the front page's **Start here** box. The front page has the same list, folded away under **Every screen, with its address**.

On a projector or TV, no PIN: `ADDRESS/s/watch`, then the card for the display. Chapter 5 lists the five. **On the video switcher's overlay input:** `ADDRESS/s/program` (chapter 4).

Graphics for OBS or a video switcher

SCREEN	URL	NOTES
Program overlay	`ADDRESS/s/program`	The scoreboard. See-through 1920×1080 in OBS; graphics on black by itself in a browser, for a switcher's luma key, with Cheesy's overlay behind it when the desk stops (chapter 4)
Telestrator render	`ADDRESS/s/tele`	See-through over the replay in OBS; ink on black on its own
Arcade overlay	`ADDRESS/s/arcade`	Over the game capture in OBS; on black on its own
Trivia overlay	`ADDRESS/s/trivia`	Crowd game. A full picture with its own background: cut to it rather than laying it over a camera

Run the show: these need the PIN

SCREEN	URL	WHO USES IT
Desk console	`ADDRESS/s/desk`	Desk manager
Talent view	`ADDRESS/s/talent`	On-air talent
Replay console	`ADDRESS/s/replay`	Replay operator
Telestrator pad	`ADDRESS/s/draw`	Analyst's tablet
Arcade console	`ADDRESS/s/arcadedesk`	Arcade scorer
Trivia host	`ADDRESS/s/triviadesk`	Trivia host
House audio player	`ADDRESS/s/house`	**On the music machine only**
Judge Advisor: awards	`ADDRESS/s/awards`	JA only. Takes the awards code, not the desk PIN
Head referee review	`ADDRESS/s/var`	Head referee. Read-only
Phone remote	`ADDRESS/s/remote`	Anyone away from the desk

Before the event

SCREEN	URL	WHO USES IT
Team media	`ADDRESS/s/media`	Whoever has the robot photos
Post-match cards	`ADDRESS/s/cards`	Social media
Event settings	`ADDRESS/s/setup`	The content lead, with the settings code

For everyone: no PIN

SCREEN	URL	NOTES
Pick a screen	`ADDRESS/s/watch`	Every display in the venue: pick the card that says *For ...*, then F11. Hand this to anyone setting up a monitor
Screen test card	`ADDRESS/s/testcard`	Full-screen it on a TV, projector or LED wall before you trust it. Chapter 5
Side screen	`ADDRESS/s/side`	Venue TVs, full screen; easiest through Pick a screen. In OBS, a full picture to cut to
When do we play?	`ADDRESS/s/next`	Parents, on their own phones
Trivia play	`ADDRESS/s/quiz`	The crowd's phones
Shout-outs	`ADDRESS/s/gp`	Submit a Gracious Professionalism moment. Moderated at the desk

Appendix B · The Friday setup list

Print this. Tick it off.

AV rack

- [ ] Desk laptop on the field network, plugged into power
- [ ] `CalGamesContentDesk.exe` copied across and run once
- [ ] Address and PIN written on the whiteboard
- [ ] Front page: *Field: connected*, and the **Switcher** line names the ATEM (chapter 9)
- [ ] Switcher's overlay computer on `/s/program`, full screen; the front page's **Screens** line has nothing to fix
- [ ] With that page up, stop the desk (or pull its cable) for 5 seconds: the input shows Cheesy Arena's score on black, then the desk's graphics again once it is back

OBS (skip this group if the event does not stream through OBS)

- [ ] OBS installed
- [ ] Six scenes created, names checked twice (chapter 4, step 2)
- [ ] Program overlay added to `CG_MATCH`, both boxes unticked
- [ ] Overlay pasted *as reference* into `CG_SCORE`, `CG_REPLAY` and `CG_DESK`
- [ ] `CG Replay Clip` Media Source in `CG_REPLAY`, under the Overlay (chapter 4, step 4b)
- [ ] `"obs"` set in `config.json` if anyone will take replays to program (chapter 4, step 7)
- [ ] Stream key entered, test stream run and stopped
- [ ] Recording tested: record thirty seconds and play it back

Screens

- [ ] Test card checked on every display: corners visible, circle round, clock ticking
- [ ] Program full screen on the projector (`/s/watch`, **Program**, F11): the CalGames backdrop or the match, never a white page
- [ ] Side screens up, full screen, rotating
- [ ] Pit monitors up
- [ ] Trivia monitor up, QR readable from where the crowd stands

Audio

- [ ] Music machine identified and `/s/house` open on it
- [ ] Announcer mics tested through the stream, listening on a phone
- [ ] Confirmed music does **not** appear on the stream

Practising without leaving a trace

The desk rebuilds the day from the day's event log every time it starts. A practice run on the event machine therefore comes back as if it really happened: practice cards on the announcer's ledger, matches nobody played in the coverage report, and a practised award reveal ticking that award off as presented, to a team that did not win it.

START-PRACTICE.cmd is already safe: the pretend match it runs is tagged, and the desk sets its log aside. What needs the flag is a rehearsal you drive by hand on the real desk, which looks identical to the show. Start it with:

```
START-DESK.cmd /rehearsal
```

Everything behaves exactly as it does on the day; only the log is set aside. The **Start here** box and the Doors check say so while it is running. To go back to normal before doors, **end the desk in Task Manager first** (Ctrl+Shift+ Esc, **CalGames 2026 Content Desk**, End task), then double-click the exe: a double-click while it is still in practice mode pops a warning box saying so and how to end it, then reopens the practice desk.

People

- [ ] Everyone has read their chapter
- [ ] Desk manager has run a practice match end to end, started in practice mode so it does not become the day's history (see below)
- [ ] Desk manager has run one match with the field disconnected, on purpose
- [ ] Talent has seen the talent view and knows what an outlined number means

Content

- [ ] Robot photos uploaded (`/s/media`)
- [ ] Sponsors entered (`/s/setup` , settings code)
- [ ] Run of show entered (`/s/setup` , settings code)
- [ ] Event name, year, and TBA key checked (`/s/setup` , settings code)
- [ ] All three codes changed from the shipped defaults; settings code given to the content lead
- [ ] Walk-up songs in place for playoff introductions
- [ ] Award list checked against this year's ceremony scripts, both days, in running order (the JA's page, `/s/awards`). The twelve 2026 awards ship with the desk; confirm the wording and the order before doors.
- [ ] Awards code set and given to the Judge Advisor (and to nobody else); JA shown their page (`/s/awards`)
- [ ] Recognition and info slides entered (setup crew names, session times)

Appendix C · Words this guide uses

WORD	MEANS
Browser Source	An OBS source that shows a web page. How the graphics get into OBS, at an event that uses it
Full screen	F11 in Chrome or Edge. How every venue display shows its screen without OBS
Scene	One arrangement of pictures in OBS. You switch between them
Overlay	The graphics on top of the camera picture: score bar, clock, names
Lower third	The name strip along the bottom
Program	What the audience is seeing right now
Cut	Switch instantly from one scene to another
On deck	The match after the one being played
Surrogate	A team playing an extra match that does not count for their record
Luma key	A way hardware switchers lay graphics over video, using black as transparent
Switcher	The video switcher (on this kit, an ATEM) that cuts the cameras and keys the graphics over them
Doors	When the audience is let in. The deadline the pre-show check exists for

This guide is part of the CalGames Content Desk, which is open source under the Apache 2.0 licence. Corrections are welcome and genuinely useful: if something here was wrong or confusing while you were standing in a gym trying to make it work, that is a bug in this document.